



Paved road in the ruins of the Roman city of Pompeii



In his book *Elements*, written around 300 BCE, Greek mathematician Euclid established the basis of geometry for the next 2,000 years.

50 BCE

### Glassblowing in Syria

Roman glassblowers in the eastern province of Syria discovered that a more even flow of molten glass could be achieved by blowing it through a thin tube. This created higher-quality and stronger glass, so vessels could be made in more complex shapes and lasted longer.



Roman blown-glass containers in the shape of doves, 1st century CE

420 BCE

### Naming atoms

Early Greek philosophers and scientists thought hard about what basic substance made up the Universe. Democritus of Abdera proposed that all matter consisted of tiny particles that could not be divided, which he called atoms, the Greek word for "uncuttable" (see pp.168–169).

312 BCE

### First Roman road

The Romans built a huge network of roads, beginning with the Via Appia. Its construction started in 312 BCE, and the road connected Rome to the southern Italian city of Capua. The roads were generally built on clay beds filled with loose gravel, and were topped with paving stones or cobbles. The high quality of the Roman roads greatly speeded up communications within the Roman Empire.

400

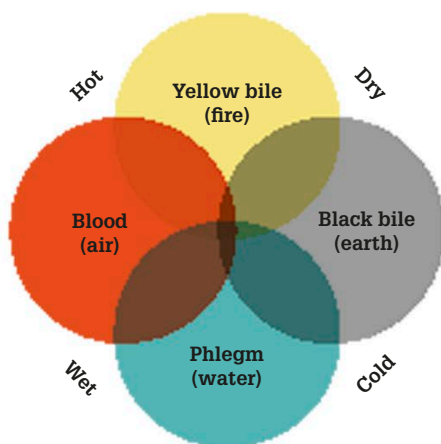
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400 BCE

### The four humors

Greek physician Hippocrates developed the idea that the body has four basic substances, or "humors": blood, phlegm, black bile, and yellow bile. Hippocrates taught that illness was caused when the humors were out of balance, a theory proved to be incorrect.



People's health was said to depend on their mix of humors.



200 BCE

Ancient Chinese compass with a magnetized iron spoon as a pointer

### Magnetic lodestone

The Chinese were the first to describe lodestone, a naturally occurring magnet. They saw that rubbing lodestone against iron magnetizes the iron. This enabled them to create primitive compasses in which an iron ladle or spoon pointed north.

c 100 BCE

### The Antikythera

The Antikythera mechanism is a complicated ancient device with toothed dials. It was discovered in a shipwreck in 1900 and is thought to be around 2,000 years old. The mechanism has more than 30 gears, and was probably used to calculate the positions of astronomical objects and to predict eclipses of the Sun and Moon.



Remains of the Antikythera mechanism